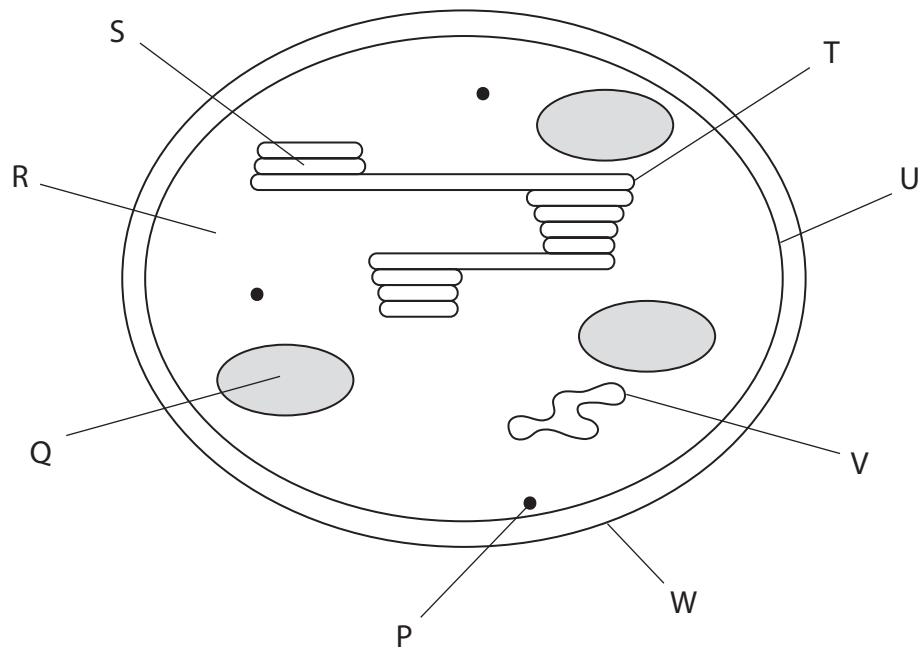


1 The structure of a chloroplast is related to its role in photosynthesis.

The diagram shows a chloroplast.



(a) Which is the site of photolysis?

(1)

- ☐ **A** P
- ☐ **B** R
- ☐ **C** T
- ☐ **D** V

(b) Which contains a gene coding for RUBISCO?

(1)

- ☐ **A** P
- ☐ **B** Q
- ☐ **C** S
- ☐ **D** V

(c) Where do hydrogen ions accumulate for use in photosynthesis?

(1)

- ☐ **A** Q
- ☐ **B** R
- ☐ **C** S
- ☐ **D** V

(d) (i) How many of the structures labelled P, T, U and W are made of phospholipids?

(1)

- ☐ **A** 1
- ☐ **B** 2
- ☐ **C** 3
- ☐ **D** 4

(ii) Which describes a phospholipid?

(1)

- ☐ **A** glycerol attached to two fatty acids and one phosphate group
- ☐ **B** glycerol attached to two fatty acids and two phosphate groups
- ☐ **C** glycerol attached to three fatty acids and one phosphate group
- ☐ **D** glycerol attached to three fatty acids and two phosphate groups

(e) The length of a cell was measured. It was found to be 0.0085 cm long.

The ratio of the length of this cell to the length of a chloroplast was 11.7:1.

Calculate the length of this chloroplast.

Give your answer in μm to 2 significant figures.

(2)

Answer..... μm

(Total for Question 1 = 7 marks)